

Evaluation of Adaptive Optics for Correcting Atmospheric Distortions in Free-Space Optical Communication

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This study evaluates the performance of an adaptive optics (AO) system in correcting atmospheric wavefront distortions for Free-Space Optical (FSO) communication. Experimental results show that the AO system significantly improves signal intensity and SNR within moderate turbulence conditions, with SNR enhancement observed between 136 ± 10 J and 450 ± 70 J. However, performance declines under extreme turbulence due to physical and computational constraints. Despite these limitations, AO systems demonstrate the potential to enhance FSO communication by improving signal stability, reducing retransmissions, and enabling higher data rates.

I. INTRODUCTION

INTRODUCTION

Free-Space Optical (FSO) communication [1] is a promising next-generation wireless, point-to-point communication technology. Utilising low-Earth orbit (LEO) satellites and infrared lasers, FSO offers significantly higher bandwidth compared to current 5G networks, along with broader coverage [2]. Despite these advantages, the technology faces significant challenges, primarily wavefront distortion caused by atmospheric density variations [3].

Atmospheric turbulence, arising from temperature gradients in the air, induces turbulent eddies that alter the refractive index, leading to optical wavefront distortion [4]. These distortions degrade signal quality by increasing instability, which manifests as a higher signal-to-noise ratio (SNR). As a result, FSO systems often require retransmission of data to ensure accurate reception by ground-based receivers, effectively reducing bandwidth and network speed. Furthermore, the signal instability limits the feasibility of advanced multiplexing techniques, constraining the achievable data rates.

(AO) system is a potential solution [6]. An AO system typically consists of three main components: a wavefront sensor (WFS), a deformable mirror (DM), and a processing unit. The WFS detects wavefront distortions based on a reference signal and relays digitised data to the processing unit. The processor computes the Zernike polynomials [7] representing the distortions and adjusts the DM to generate an anti-phase surface. This corrective action neutralises the distorted wavefront, producing an undistorted output signal. The corrected wavefront is fed back into the WFS, enabling iterative refinement through a feedback loop.

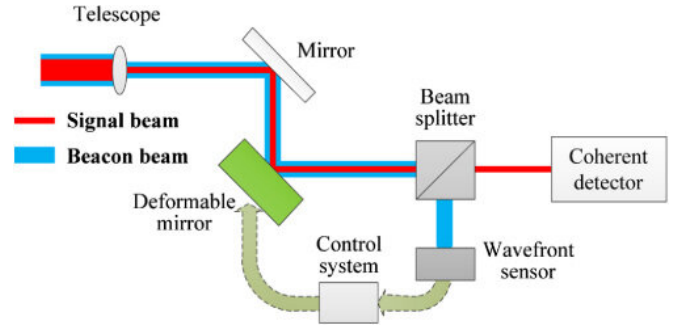


FIG. 2: Closed Loop AO System [8].

AO systems are particularly effective in correcting common atmospheric distortions such as coma, astigmatism, defocus, and tip/tilt. However, their performance is limited in certain scenarios [9]:

- **Strong turbulence:** DM actuators have finite extension and contraction limits, which restrict correction for extreme distortions.
- **Complex wavefronts:** The finite number of DM actuators reduces the resolution of the corrective surface, leaving residual distortions.
- **Dynamic distortions:** Rapidly changing wavefronts may exceed the processing speed for Zernike polynomial calculations and DM adjustments.

This report aims to evaluate the effectiveness of an AO system in mitigating atmospheric distortions for FSO communication and to identify its limitations under varying turbulence conditions. To achieve this, simulations are performed for different turbulence intensities, and the total signal intensity and SNR are compared with and without AO correction.

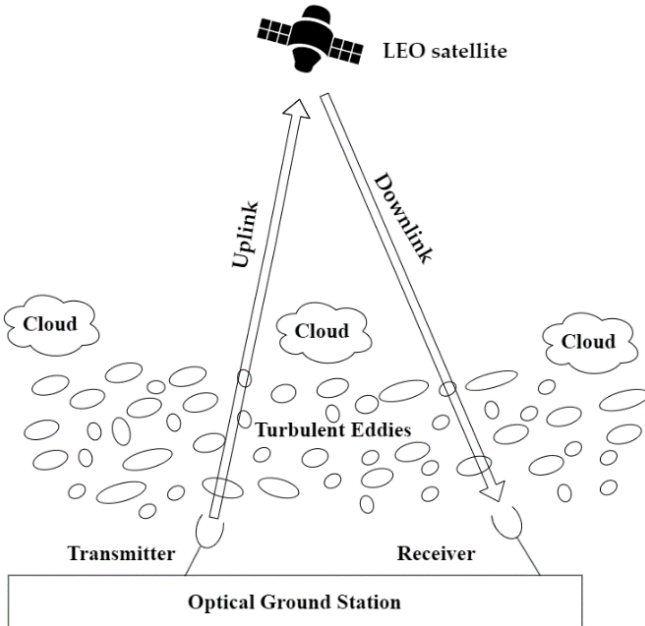


FIG. 1: Atmospheric effects on satellite-ground FSO communications for both uplink and downlink channels. [5].

To mitigate these issues, a *closed-loop adaptive optics*

The **signal** is defined as the average intensity within a specific region of interest (RoI) over time:

$$I_{\text{avg}} = \frac{\sum I_{\text{within RoI}}}{n}. \quad (1)$$

The **noise** is characterised by the standard deviation of intensity values in the RoI:

$$\text{Noise} = \sigma_{I_{\text{within RoI}}}. \quad (2)$$

The SNR is then calculated as the ratio of signal intensity to noise, with a rolling average used to analyse trends:

$$\text{SNR} = \frac{I_{\text{avg}}}{\text{Noise}}. \quad (3)$$

The *failing point* of the AO system is defined as the condition where the SNR with the AO system becomes equal to or lower than the SNR without it.

II. METHOD

A. Experimental Setup

The experimental setup was designed to evaluate the performance of a closed-loop adaptive optics (AO) system under simulated atmospheric turbulence. The system includes a heating wire to simulate turbulence, a deformable mirror (DM) for wavefront correction, and a CCD camera to record the corrected interference pattern.

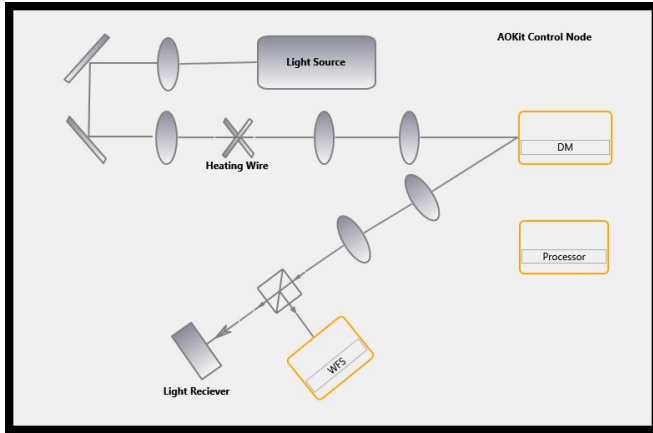


FIG. 3: A closed-loop AO system with a heating wire and a CCD camera.

A laser beam of wavelength 600 nm propagates through a turbulent region created by a heating wire, reflects off the DM, and is then split by a beam splitter. One split beam is directed to a CCD camera (frame rate: 34.815 fps) for measurement, while the other is sent to the wavefront sensor (WFS) as feedback. The processor, connected to the WFS and DM, calculates up to 36 Zernike polynomials using Thorlabs AO Kit software [10] and adjusts the DM actuators to correct wavefront distortions. Two $4f$ relay systems [11] ensure optical alignment, maintaining all key components on the same optical plane.

1. Alignment and Systematic Error Mitigation

To minimise systematic errors:

1. Component heights were aligned using a ruler, ensuring the laser beam passed through the centre of each component.
2. The laser was adjusted to propagate parallel to the optical table.
3. Collimation was verified at each $4f$ relay system.
4. The WFS and CCD camera were positioned at the focal points of the final lenses for accurate image capture.
5. The WFS tilt was adjusted to remain perpendicular to the incident beam.
6. The heating wire was placed on a thermal insulator to prevent heat transfer to optical components, avoiding material expansion.

2. Calibration and System Validation

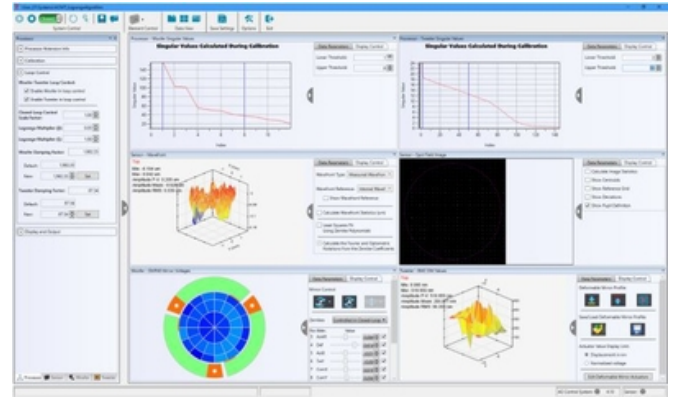


FIG. 4: Thorlabs Adaptive Optics Kits Software [10]

The laser source was activated, and the DM was calibrated using Thorlabs AO Kit software. Calibration involved initiating the closed-loop system and ensuring the root mean square (RMS) wavefront error after DM correction remained below 1 RMS.

3. Simulation of Atmospheric Turbulence

Turbulence was simulated using a 3.1Ω heating wire powered by a DC supply delivering 0.9 A. Measurements were recorded with the following devices:

- A digital multimeter (resolution: 0.1Ω) for resistance.
- An analogue display (resolution: 0.1 A) for current.

The power dissipated by the heating wire was calculated as:

$$P = I^2 R$$

Assuming all dissipated energy heated the surrounding air:

$$\Delta Q = P \Delta t$$

This heat transfer increased turbulence, inducing phase distortions in the laser beam.

4. Experimental Procedure

The CCD camera was set to an exposure time of 1.013 ms, avoiding sensor saturation. The procedure was as follows:

1. The CCD recorded a 10-second "reference pattern" (Airy disk without turbulence).
2. The heating wire was powered, and turbulence was allowed to develop for 5 minutes.
3. Interference patterns during this period were compared to the reference to evaluate distortion.
4. The procedure was repeated under two conditions:
 - AO system disabled (DM flat).
 - AO system enabled (DM dynamically correcting wavefront distortions).

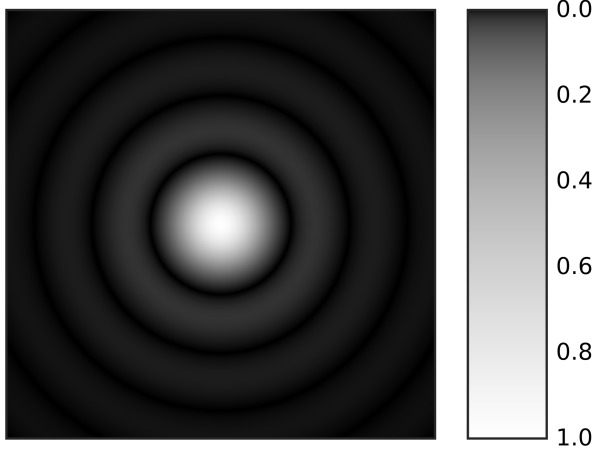


FIG. 5: Airy disk pattern [12].

B. Data Analysis

The recorded video contains frames of data with x, y coordinates on the CCD and the detected intensity values at each pixel. The frames were read as NumPy matrices using the `imageio` module [13]. The following steps were performed to quantify the signal and signal-to-noise ratio (SNR) within the region of interest (RoI).

Step 1: Defining the Region of Interest (RoI)

The first 10 seconds of the reference video (≈ 300 frames, frame rate: 34.815 fps) were extracted. The total intensity across all frames was computed as:

$$M_{\text{total}} = \sum_{i=1}^{300} M_i, \quad (4)$$

where i represents the frame count, and M_i is the intensity matrix for the i -th frame.

The resulting intensity matrix M_{total} was optimised to the 2D Gaussian function defined as:

$$G(x, y) = \text{offset} + A \exp \left(- \left[a(x - x_0)^2 + 2b(x - x_0)(y - y_0) + c(y - y_0)^2 \right] \right), \quad (5)$$

where:

$$a = \frac{\cos^2 \theta}{2\sigma_x^2} + \frac{\sin^2 \theta}{2\sigma_y^2}, \quad (6)$$

$$b = -\frac{\sin(2\theta)}{4\sigma_x^2} + \frac{\sin(2\theta)}{4\sigma_y^2}, \quad (7)$$

$$c = \frac{\sin^2 \theta}{2\sigma_x^2} + \frac{\cos^2 \theta}{2\sigma_y^2}. \quad (8)$$

Definition of Variables:

- x, y : Pixel coordinates on the CCD matrix.
- A : Amplitude of the Gaussian function (peak intensity).
- (x_0, y_0) : Centre of the Gaussian, corresponding to the location of maximum intensity.
- σ_x, σ_y : Standard deviations of the Gaussian function along the x and y axes, representing the spread of intensity in each direction.
- θ : Rotation angle of the Gaussian ellipse relative to the x -axis.
- offset : Baseline intensity level, accounting for background noise.

The optimisation was performed using the `scipy` module [14], minimising the residuals between the measured total intensity matrix M_{total} and the fitted Gaussian function $G(x, y)$.

The RoI was then defined as an ellipse centred at (x_0, y_0) , with horizontal and vertical radii equal to σ_x and σ_y , rotated by the angle θ .

Step 2: Binning the Main Video and Calculating Average Signal and Noise

The 5-minute main video was divided into bins of 100 frames (≈ 2.87 s per bin). For each bin, the total signal intensity within the defined RoI was calculated as:

$$\bar{I} = \sum_{i=1}^{100} I_i, \quad (9)$$

where I_i represents the intensity within the RoI for the i -th frame in the bin.

The noise for each bin was calculated as the standard deviation of the 100 intensity values:

$$\text{Noise} = \sigma_{I_{1:100}}. \quad (10)$$

This process was repeated for the entire video, generating signal data points $\bar{I}_1, \bar{I}_2, \bar{I}_3, \dots$ and noise data points $\sigma_1, \sigma_2, \sigma_3, \dots$. The median time of each bin was recorded as the corresponding time for the average signal and noise.

Step 3: Calculating the Rolling Average SNR

The SNR for each bin was calculated as:

$$\text{SNR}_i = \frac{\bar{I}_i}{\sigma_i}. \quad (11)$$

To smooth the trend of SNRs over time, a rolling average was applied to 11 adjacent SNR data points:

$$\text{SNR}_{\text{avg}} = \frac{1}{11} \sum_{j=i-5}^{i+5} \text{SNR}_j, \quad (12)$$

where SNR_{avg} represents the smoothed SNR, and j spans the indices of the adjacent bins. The median time of the 11 SNRs was used as the time for SNR_{avg} .

Step 4: Calculating the Percentage Difference

The above analysis was conducted for two cases: with the AO system enabled and with the AO system disabled. The percentage difference in signal and SNR between these two cases was calculated as:

$$\% \Delta I = \frac{I_{\text{with_AO}} - I_{\text{without_AO}}}{I_{\text{without_AO}}} \times 100, \quad (13)$$

$$\% \Delta \text{SNR} = \frac{\text{SNR}_{\text{with_AO}} - \text{SNR}_{\text{without_AO}}}{\text{SNR}_{\text{without_AO}}} \times 100. \quad (14)$$

The AO system's effect was categorised as:

- **Positive:** If $\% \Delta \text{SNR} > 0$.
- **Negative:** If $\% \Delta \text{SNR} < 0$.

Step 5: Converting Time to Heat

The turbulence level, represented by the additional heat introduced into the system, was calculated as:

$$\Delta Q = I^2 R \Delta t, \quad (15)$$

where I is the current through the heating wire (0.9A), R is its resistance (3.1Ω), and Δt is the time interval. The heat ΔQ served as a proxy for the degree of turbulence.

Step 6: Calculating the Intersection Q

The intersection points, representing the turbulence levels where the adaptive optics (AO) system transitions between improving and degrading the SNR, were calculated using the ΔQ values derived in Step 5. The methodology was as follows:

- Two key transitions in the SNR improvement ($\% \Delta \text{SNR}$) were identified:
 1. The first significant transition where $\% \Delta \text{SNR}$ changes from negative to positive. This point marks the onset of effective AO correction.
 2. The last significant transition where $\% \Delta \text{SNR}$ changes from positive to negative. This point marks the limit of AO system effectiveness.

- The Q values corresponding to these transitions were averaged to determine the intersection points:

$$Q_{\text{intersect}} = \frac{Q_{\text{start}} + Q_{\text{end}}}{2}, \quad (16)$$

where Q_{start} is the turbulence level at the first transition and Q_{end} is the turbulence level at the last transition.

- Fluctuations near the Q -axis, caused by noise or minor variations, were ignored by focusing on the most significant transitions. This ensured the calculated intersections represent meaningful changes in the AO system's performance.

This step provides a clear demarcation of the AO system's operational range under varying turbulence conditions, aiding in the analysis of its effectiveness for FSO communication.

RESULTS*Gaussian Optimisation*

Figures 6 and 7 show the summed intensity and the optimised 2D Gaussian function for the reference video without AO correction. Similarly, Figures 8 and 9 illustrate the same for the reference video with AO correction.

The optimised parameters for the 2D Gaussian function, as defined in Equation 5, were as follows:

Without AO:

- Amplitude: 7152.0083
- x_0 : 764.776
- y_0 : 614.056
- σ_x : 2.6383
- σ_y : 7.1786
- θ : 0.2119
- Offset: 0.5250

With AO:

- Amplitude: 52897.4846
- x_0 : 803.520
- y_0 : 718.401
- σ_x : 2.2418
- σ_y : 1.9498
- θ : 1.4694
- Offset: 0.3322

From these optimised parameters, the elliptical region of interest (RoI) was defined as:

- **Without AO:** Centred at (764.776, 614.056), with radii $\sigma_x = 2.6383$ and $\sigma_y = 7.1786$.
- **With AO:** Centred at (803.520, 718.401), with radii $\sigma_x = 2.2418$ and $\sigma_y = 1.9498$.

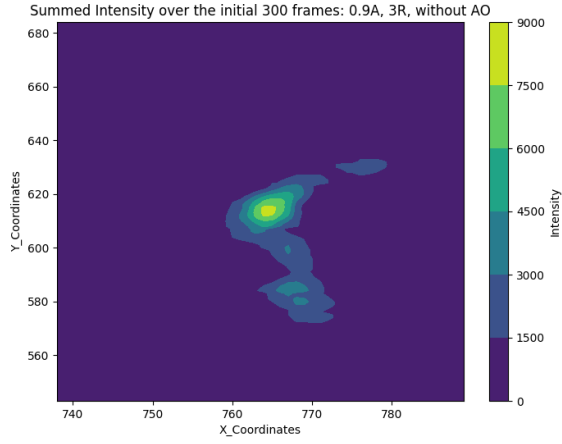


FIG. 6: Summed intensity for the reference video: without AO.

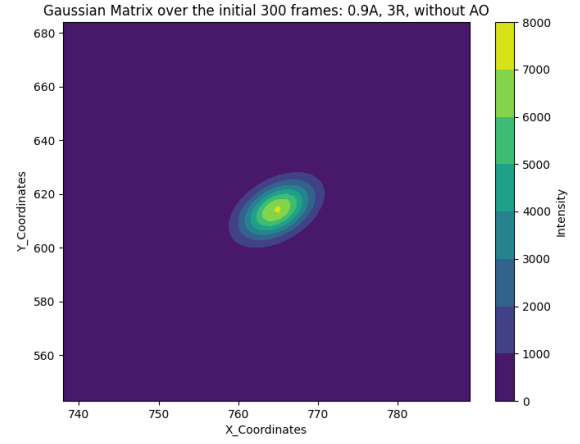


FIG. 9: Optimised 2D Gaussian function: with AO.

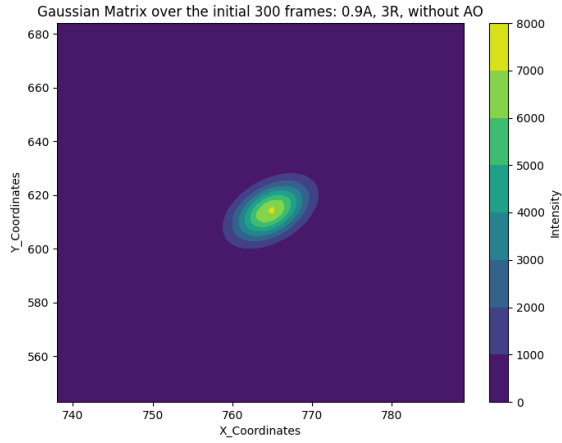


FIG. 7: Optimised 2D Gaussian function: without AO.

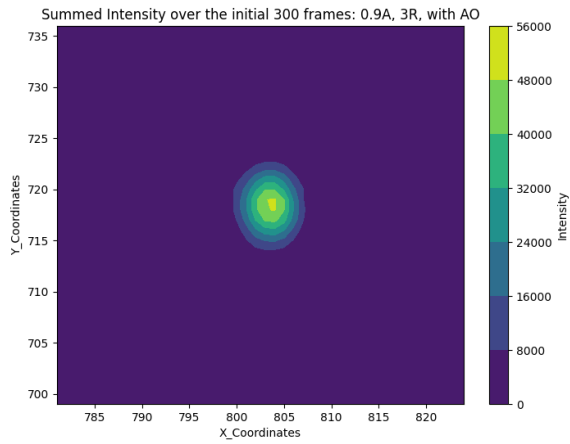


FIG. 8: Summed intensity for the reference video: with AO.

Signal Over Heat: With and Without AO

Figure 10 shows the total signal intensity as a function of the additional heat (ΔQ) with and without AO correction. Signal improvement, defined as the percentage increase in signal intensity due to AO, is illustrated in Figure 11.

- **Observation:** The signal intensity with AO correction consistently outperforms the signal without AO across most of the turbulence range.
- **Key Finding:** The AO system's effectiveness diminishes as turbulence increases beyond a critical threshold, where the signal intensities converge.

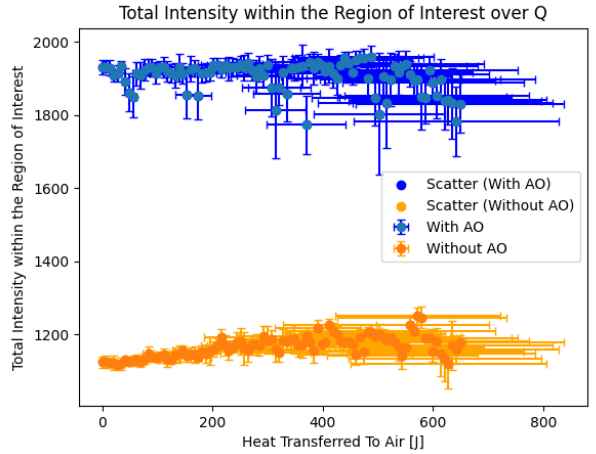


FIG. 10: Total signal intensity as a function of heat: with and without AO.

SNR Over Heat: With and Without AO

The SNR as a function of additional heat (ΔQ) is shown in Figure 12, while the SNR improvement percentage due to AO is presented in Figure 13. The green region in Figure 13 highlights where the AO system improves the FSO signal, while the red region indicates where AO worsens the distortion.

- **Improvement Range:** AO begins improving the SNR at $\Delta Q = 136 \pm 10$ J and ceases improvement at $\Delta Q = 450 \pm 70$ J.
- **Observation:** The improvement diminishes as turbulence increases, likely due to the physical and computational limits of the AO system.

DISCUSSION

The results presented demonstrate the potential of adaptive optics (AO) systems in improving Free-Space Optical (FSO) communication by mitigating the effects of atmospheric turbulence. The discussion below interprets the findings, highlights the implications for FSO systems, and identifies both the limitations and benefits of employing AO in such applications.

Implications for FSO Communication Systems

Signal Strength Enhancement

Figures 10 and 11 show that the total signal intensity within the region of interest (RoI) was consistently higher when the AO system was applied. This indicates that the AO system effectively reduces wavefront distortions caused by atmospheric turbulence, resulting in a stronger and more stable optical signal. For FSO communication systems, this implies:

- Improved signal strength translates to reduced transmission errors, minimising the need for retransmission and increasing effective bandwidth.
- The system becomes more robust against noise, particularly in urban environments or scenarios with mild to moderate turbulence.
- Higher signal intensity allows for the use of advanced modulation schemes, enabling higher data rates and more efficient utilisation of the optical spectrum.

SNR Improvement and Operating Range

Figure 13 illustrates that the AO system improves the signal-to-noise ratio (SNR) within a heat range of 136 ± 10 J to 450 ± 70 J. This corresponds to moderate levels of turbulence, where the AO system is most effective.

- The initial negative SNR improvement observed before 136 J is likely due to the absence of significant turbulence. In this scenario, the deformable mirror (DM), which is not perfectly flat, introduces minor distortions, causing light scattering. This suggests that FSO systems operating in environments with minimal turbulence may not benefit from AO and would perform better with a simple flat mirror.
- Beyond 450 J, the AO system fails to improve the SNR. This is consistent with the limitations discussed in the introduction:
 - The maximum extension of DM actuators ($13.6 \mu\text{m}$) constrains the system's ability to correct extreme distortions.
 - The processing unit's limitation to calculating only 36 Zernike polynomials reduces its capacity to address complex wavefront aberrations.
 - The finite processing speed of the AO system causes delays in responding to rapidly changing turbulence, reducing correction effectiveness in dynamic conditions.

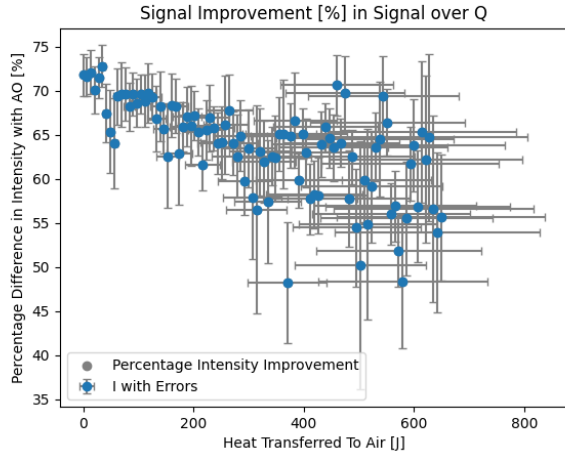


FIG. 11: Signal improvement (%) due to AO.

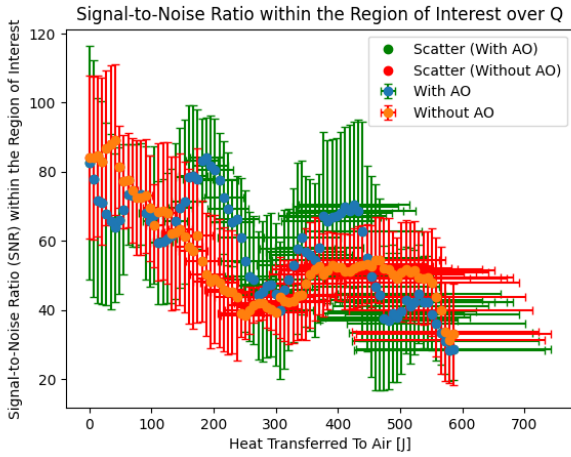


FIG. 12: SNR as a function of heat: with and without AO.

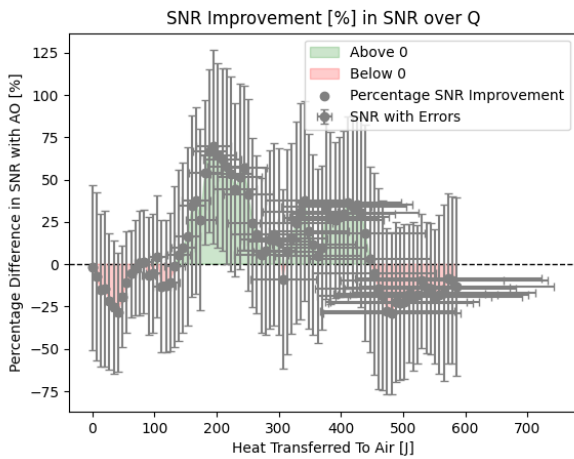


FIG. 13: SNR improvement (%) due to AO. Green region: improvement. Red region: degradation.

For FSO communication systems, this operating range highlights the importance of optimising AO system parameters based on the expected level of turbulence. Deploying AO in regions with moderate atmospheric disturbances, such as suburban areas, could maximise its benefits.

Limitations of the AO System in FSO Communication

While the AO system demonstrated significant improvements in signal strength and SNR, its limitations must be acknowledged:

- **Physical Constraints of the DM:** The maximum actuator extension restricts the correction of extreme distortions, reducing the system's effectiveness in high-turbulence scenarios.
- **Limited Polynomial Orders:** Correcting only up to 36 Zernike polynomials restricts the AO system's capacity to address higher-order aberrations commonly observed in severe atmospheric turbulence.
- **Processing Speed:** The computational latency of detecting distortions and adjusting the DM limits the system's ability to respond to rapid turbulence changes.

Systematic Errors and Potential Mitigation

The following potential systematic errors may have influenced the results:

- **Laminar Flow Instead of Turbulence:** As shown in Figure 12, fluctuations in the SNR with AO could indicate that the heating wire created a uniform uplift in the air, producing laminar flow instead of turbulence. Laminar flow is easier for the AO system to correct, artificially boosting the SNR.
 - **Mitigation:** Future experiments could use more sophisticated turbulence simulation tools, such as rotating phase screens, to produce more realistic atmospheric distortions.
- **Expansion-Induced Defocus:** The heating wire may have caused thermal expansion of the rods in the $4f$ relay, introducing additional defocus or collimation errors unrelated to turbulence.
 - **Mitigation:** Employing thermally insulated components or monitoring beam quality at intermediate stages could help isolate such errors.
- **Minor Misalignment:** Small misalignments in the optical components could have contributed to unaccounted distortions, slightly altering the results.
 - **Mitigation:** Incorporate automated alignment tools to reduce human error during setup.

Benefits of Using AO Systems in FSO Communication

Despite its limitations, the AO system offers several advantages for FSO communication:

- **Improved Bandwidth Utilisation:** By reducing the need for retransmission, AO enables higher throughput and more efficient bandwidth usage.
- **Enhanced Signal Stability:** The system mitigates turbulence-related signal degradation, ensuring more reliable communication links over long distances.
- **Applicability to Advanced Modulation Schemes:** With improved SNR, AO systems make it feasible to employ high-order modulation schemes, significantly increasing data rates.
- **Scalability to Multiple Channels:** AO systems can be adapted for multi-channel FSO communication, enabling simultaneous data streams across separate wavelengths.

Future Work and Recommendations

To maximise the potential of AO systems in FSO communication, future work should focus on:

- Developing faster processors to handle higher-order distortions and dynamic turbulence more effectively.
- Exploring hybrid correction methods, combining AO with machine learning algorithms for adaptive real-time optimisation.
- Implementing more realistic turbulence simulators to test system performance under diverse environmental conditions.

CONCLUSION

This study evaluated the performance of an adaptive optics (AO) system in mitigating atmospheric wavefront distortions for Free-Space Optical (FSO) communication. By analysing signal intensity and signal-to-noise ratio (SNR) under simulated turbulence, the experiment demonstrated both the potential and limitations of AO systems in enhancing FSO communication.

The results showed that AO systems significantly improve FSO signal quality under moderate turbulence conditions. Figures 10 and 11 revealed that the total signal intensity within the region of interest (RoI) was consistently higher with AO correction, leading to:

- Stronger and more stable signals, reducing the likelihood of transmission errors.
- Improved bandwidth utilisation by minimising retransmissions and enabling high-order modulation schemes for faster data rates.
- Enhanced system reliability over longer communication distances.

The SNR improvement observed between 136 ± 10 J and 450 ± 70 J (Figure 13) highlights the optimal operational range for AO systems. Within this range, the AO system effectively adapts to atmospheric turbulence, improving communication stability. However, at lower turbulence levels (< 136 J), the AO system introduced minor distortions due

to the non-flat surface of the deformable mirror (DM). Beyond 450 J, the AO system failed to correct the distortions, constrained by:

- The maximum actuator extension of the DM (13.6 μm).
- The limited calculation of 36 Zernike polynomials, which is insufficient for higher-order wavefront distortions.
- Processing delays in dynamic turbulence, where rapid changes outpace the correction response.

Implications for FSO Communication

These findings underline the critical role of AO systems in improving FSO communication under moderate turbulence conditions. By enhancing signal strength and SNR, AO systems support:

- The deployment of FSO communication in suburban or urban areas, where turbulence levels are typically within the AO system's operational range.
- The integration of advanced modulation techniques, such as quadrature amplitude modulation (QAM), for higher spectral efficiency.
- The expansion of FSO communication networks to bridge long distances with minimal signal degradation.

However, in extreme turbulence scenarios, AO systems require further optimisation to address their physical and computational constraints. Similarly, environments with minimal turbulence may not benefit from AO correction, as the introduction of minor distortions can outweigh the advantages.

Future Outlook

Advancing AO systems for FSO communication requires innovation in both hardware and software. Future developments should focus on:

- Designing deformable mirrors with greater actuator extension and higher resolution to handle severe wavefront distortions.
- Increasing computational capacity to calculate higher-order Zernike polynomials and improving processing speeds to correct dynamic turbulence in real time.
- Developing hybrid correction methods that integrate AO with machine learning algorithms. Machine learning can predict distortion patterns and guide AO adjustments, enhancing correction accuracy and speed.
- Conducting experiments under real-world atmospheric conditions, such as urban heat islands or varying weather patterns, to validate AO system performance in practical deployments.

Final Remarks

Adaptive optics systems are a promising solution for improving the reliability and performance of FSO communication. While their effectiveness is currently limited by physical and computational constraints, their ability to enhance signal stability, increase SNR, and support high-speed data transmission makes them a critical technology for the next generation of optical communication networks. Continued advancements in AO systems will enable their broader adoption in FSO communication, bridging the gap between theoretical potential and practical implementation.

REFERENCES

- [1] Touati, A., Touati, F., Abdaoui, A., Khandakar, A., Bouallègue, A. (2018). Prospects and Challenges of Free Space Optical Communications. DOI: 10.1007/978-3-030-02849-7_4.
- [2] Kaushal, H., Kaddoum, G. (2017). Optical Communication in Space: Challenges and Mitigation Techniques. DOI: 10.1109/COMST.2016.2603518.
- [3] Chan, V. (2006). Free-Space Optical Communications. DOI: 10.1109/JLT.2006.885252.
- [4] Kaushal, H., Kaddoum, G. (2017). Optical Communication in Space: Challenges and Mitigation Techniques. DOI: 10.1109/COMST.2016.2603518.
- [5] Atmospheric Effects on Satellite-Ground Free Space Uplink and Downlink Optical Transmissions - Scientific Figure on ResearchGate. Available from: researchgate.net
- [6] Guiomar, F., Fernandes, M., Nascimento, J. L., Rodrigues, V., Monteiro, P. (2022). Coherent Free-Space Optical Communications: Opportunities and Challenges. DOI: 10.1109/jlt.2022.3164736.
- [7] Pham, H., Dang, N., Vu, L. T., Bui, H. T. (2014). A Survey of Performance Improvement Methods for Free-Space Optical Communication Systems. DOI: 10.1109/ATC.2014.7043491.
- [8] Slant-path coherent free space optical communications over the maritime and terrestrial atmospheres with the use of adaptive optics for beam wavefront correction - Scientific Figure on ResearchGate. Available from: researchgate.net
- [9] Kaushal, H., Kaddoum, G. (2017). Optical Communication in Space: Challenges and Mitigation Techniques. DOI: 10.1109/COMST.2016.2603518.
- [10] Thorlabs - Photonics Products solutions. Manual Download Page
- [11] Kang, H., Ye, J., Jahannia, B., Altaleb, S., Wang, H., Heidari, E., Asadizanjani, N., Sorger, V., Dalir, H. (2024). Advanced 4f-based free-space optical system. DOI: 10.1117/12.3003237.
- [12] By Sakurambo at English Wikipedia. Public Domain: <https://commons.wikimedia.org/w/index.php?curid=18640280>
- [13] Imageio Contributors. "Imageio Documentation." Public Domain: <https://imageio.readthedocs.io/en/stable/> Accessed: January 22, 2025.
- [14] Virtanen, Pauli, et al. "SciPy: Fundamental Algorithms for Scientific Computing in Python." *Nature Methods*, vol. 17, 2020, pp. 261–272. DOI: <https://doi.org/10.1038/s41592-019-0686-2> Accessed: January 22, 2025.

ERROR APPENDIX

ERROR PROPAGATION

Accurate estimation of uncertainties is essential for reliable experimental results. This section outlines the propagation of errors in calculating heat (Q), percentage differences, and intersection points, referencing the equations and methods used in the main report.

Sources of Error

The primary sources of error in the experiment include:

- **Frame Rate (fps):** The CCD camera's frame rate introduces an uncertainty of $\delta \text{fps} = 0.001 \text{ s}^{-1}$, based on its digital resolution.
- **Current (I):** The applied current has an uncertainty of $\delta I = 0.05 \text{ A}$, derived from half the ammeter resolution.
- **Resistance (R):** The heating wire's resistance has an uncertainty of $\delta R = 0.1 \Omega$, determined by the digital multimeter's resolution.

Heat Calculation

The heat transferred to air is calculated using:

$$Q = P \cdot t, \quad \text{where } P = I^2 \cdot R \quad (17)$$

Error in Power (P)

The error in power, δP , is propagated as:

$$\delta P = P \cdot \sqrt{\left(\frac{\delta I^2}{I^2}\right)^2 + \left(\frac{\delta R}{R}\right)^2} \quad (18)$$

where $\delta I^2 = 2 \cdot I \cdot \delta I$.

Error in Time (t)

The time, t , is calculated from the frame rate and number of frames:

$$t = \frac{\text{\#frames}}{\text{fps}} \quad (19)$$

The associated error, δt , is:

$$\delta t = \frac{\text{\#frames} \cdot \delta \text{fps}}{\text{fps}^2} \quad (20)$$

Error in Heat (Q)

The total error in heat, δQ , is:

$$\delta Q = Q \cdot \sqrt{\left(\frac{\delta P}{P}\right)^2 + \left(\frac{\delta t}{t}\right)^2} \quad (21)$$

Percentage Differences

The percentage difference between AO and non-AO cases is:

$$\% \Delta X = \frac{X_{\text{AO}} - X_{\text{No_AO}}}{X_{\text{No_AO}}} \times 100 \quad (22)$$

The error in percentage difference is propagated as:

$$\delta(\% \Delta X) = \% \Delta X \cdot \sqrt{\left(\frac{\delta X_{\text{AO}}}{X_{\text{AO}}}\right)^2 + \left(\frac{\delta X_{\text{No_AO}}}{X_{\text{No_AO}}}\right)^2} \quad (23)$$

Intersection Points

Intersection points where SNR improvement changes sign are calculated as:

$$Q_{\text{intersect}} = \frac{Q_i + Q_{i+1}}{2} \quad (24)$$

The associated error is:

$$\delta Q_{\text{intersect}} = \frac{\sqrt{\delta Q_i^2 + \delta Q_{i+1}^2}}{2} \quad (25)$$

Key Takeaways

1. The dominant source of error in heat calculations is the uncertainty in power (P), primarily due to the squared dependence on current (I). 2. Error propagation for percentage differences amplifies uncertainties, necessitating high measurement precision for reliable results. 3. Errors in intersection points are influenced by adjacent heat uncertainties, reflecting the sensitivity of results near transition ranges.

Future Improvements

To reduce uncertainties:

- Use higher-resolution instruments for current and resistance measurements.
- Incorporate real-time monitoring of frame rate and experimental conditions to minimise systematic errors.

USE OF GENERATIVE AI

I have used a generative AI service called "Wordtune" (<https://www.wordtune.com>). The service was used to spot Grammar and spelling mistakes, refining sentences for better readability. The following is to be considered: the work presented in the document contains original contributions from the author. This is to mean that I was only supposed to use "Wordtune" as a way to supplement the quality of language in the document to ensure it is sufficient enough and meets the high standards of academic writing. This statement is to ensure transparency in the usage of artificial intelligence for academic efforts and to testify that the work presented is authentic.

LAY SUMMARY

Free-Space Optical (FSO) communication is a cutting-edge technology that uses infrared lasers to transmit data wirelessly between satellites and ground stations. It offers ultra-fast data speeds and global coverage, far exceeding current 5G capabilities. However, a significant challenge to its widespread adoption is atmospheric turbulence, which distorts laser beams, reducing signal quality and reliability.

This study explores the use of adaptive optics (AO) systems as a solution to mitigate these distortions. AO systems dynamically correct wavefront errors using a deformable mirror, a wavefront sensor, and a processor that works in real-time. Simulations were conducted to evaluate the performance of AO under different turbulence levels.

Results show that AO systems significantly enhance signal strength and stability under moderate turbulence, with improvements in signal quality and reliability observed up to a critical turbulence threshold. Beyond this limit, the system's performance declines due to hardware and computational constraints.

These findings highlight the potential of AO systems to improve FSO communication, reducing transmission errors and enabling faster, more reliable data transfer. This research lays the groundwork for further innovation, with implications for satellite communication and global internet connectivity.